

ASSIGNMENT 7

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Creating and Fixing Mode Collapse in a GAN

Objective

The main objective of this assignment is to understand the problem of mode collapse in Generative Adversarial Networks (GANs) and observe how changes in the training setup can affect the quality and diversity of generated images.

1. Introduction

In this assignment, a DCGAN was implemented using the Fashion-MNIST dataset. First, the Generator was intentionally modified to make the GAN training unstable. The generated results were then observed. After that, label smoothing was introduced as an improvement technique, and the model was trained again to compare the results.

2. Creating an Unstable GAN

The GAN was intentionally made unstable by applying the following changes:

- The capacity of the Generator was reduced.
- Batch Normalization was removed from the Generator.
- The learning rate was increased to **0.01**.
- The modified GAN was trained for **18 epochs**.

These modifications affected the balance between the Generator and Discriminator and resulted in less stable training.

Training Result

The unstable GAN successfully completed 18 epochs. However, the generated samples contained noisy patterns and some outputs had similar characteristics, showing reduced image quality and diversity.

Broken Output

Broken Output Screenshot



Insert the broken GAN screenshot here.

3. Problem Observed

After training the unstable model, the generated Fashion-MNIST samples were not consistently clear. Some images contained noise, while certain samples looked similar to one another. This indicated that the model was not generating a sufficiently diverse range of samples.

The experiment did not produce complete mode collapse, but it demonstrated the effects of unstable GAN training and reduced diversity.

4. Technique Used to Improve the GAN

Label Smoothing

Label smoothing was selected to improve the stability of the GAN.

Normally, real images are assigned a target label of **1.0**. In this experiment, the real-image label was changed to **0.9**.

This prevents the Discriminator from becoming excessively confident and can help maintain a more balanced training process between the Generator and Discriminator.

The improved model was trained again for **18 epochs**.

5. Improved Output

After applying label smoothing, the GAN generated samples with better structure compared with the unstable version. The outputs also showed improved diversity in several samples.

Improved Output



Insert the improved GAN screenshot here.

6. Comparison of Results

Feature	Unstable GAN	Improved GAN
Generator	Reduced capacity	More stable training
Batch Normalization	Removed	Used in improved configuration
Learning Rate	0.01	Improved configuration
Improvement Technique	None	Label smoothing
Image Quality	Noisy and unclear	Better structured
Diversity	Limited in some samples	Improved diversity
Training Stability	Less stable	More stable

7. Explanation

1. What modification caused the model to perform poorly?

The Generator capacity was reduced, Batch Normalization was removed, and the learning rate was increased to 0.01. These modifications disturbed the training balance between the Generator and Discriminator and resulted in lower-quality generated images.

2. What did the generated images look like?

The unstable model produced Fashion-MNIST samples that were sometimes noisy and difficult to identify. Some samples had similar patterns, which indicated reduced diversity.

3. Which technique did you use to improve the model?

Label smoothing was used to improve the GAN. Instead of using **1.0** as the target label for real images, **0.9** was used. This reduces the confidence of the Discriminator and can make GAN training more stable.

4. Did the solution improve the output? Why?

Yes, the output improved to some extent after applying label smoothing. The generated samples showed better structure and diversity compared with the unstable model. Although the results were not perfect, the improved training produced more recognizable samples.

8. Conclusion

This experiment demonstrated how GAN performance can be affected by changes to the Generator architecture and learning rate. Reducing the Generator capacity, removing Batch Normalization, and using a high learning rate resulted in unstable training. Applying label smoothing helped improve the stability of the training process and produced better-quality generated samples.